

Math 243: Homework 03

Due Friday, September 11

Problem 1. Consider

$$\mathbf{r}(t) = (\cos(2t), \sin(2t)), \quad 0 \leq t \leq 2\pi.$$

This parameterization traces out the unit circle, whose circumference is 2π , but $\int_0^{2\pi} |\mathbf{r}'(t)| dt = 4\pi$. Explain why there is no contradiction. What does the integral measure in this example?

Problem 2. Consider $\mathbf{r}(t) = (t^3, t^2)$.

- (a) Compute the velocity at time $t = 0$. Why does the direction formula $\mathbf{r}'(t)/|\mathbf{r}'(t)|$ fail at $t = 0$?
- (b) Convert the curve to Cartesian form by eliminating the variable t and sketch the curve. Explain geometrically what goes wrong.

Problem 3 (Converting parametric equations to Cartesian form). Consider the curve given by the parametric equations

$$x(t) = \frac{t-2}{t+1}, \quad y(t) = \frac{t}{t-1}$$

where $-\frac{1}{2} \leq t \leq \frac{1}{2}$.

- (a) Find a Cartesian equation for the curve. Sketch a graph, and show the direction of travel. (*Hint: be careful about precisely which values of x your equation holds for.*)
- (b) How does your answer change if we specify $-1 < t < 1$? Plot the graph in that case as well. (*Note that we can't take $t = -1$ or $t = 1$ because that would result in division by zero!*)

Problem 4 (Arc length of parameterized curve). Compute the arc length of the parameterized curve:

$$\begin{cases} x(t) = \frac{4}{3}t^3 + t - 2 \\ y(t) = \frac{1}{27}(9t^2 + 2)^{\frac{3}{2}} \end{cases}$$

for $0 \leq t \leq 1$. (Don't forget to use the chain rule!)

Problem 5 (Arc length - 3d). Find the length of the following curve in 3d space:

$$\mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} 1 \\ t^2 \\ t^3 \end{bmatrix}, \quad 0 \leq t \leq 9.$$

Problem 6 (Parametric equations - tangent lines). Let C be the curve defined by the parametric equations

$$\begin{cases} x(t) = t^3 - 12t \\ y(t) = 10 - t^2 \end{cases} \tag{1}$$

for $0 \leq t \leq 3$.

Here's a plot of the graph of this curve: <https://www.desmos.com/calculator/bvti5ck5r0>

- (a.) Find an equation for the tangent line to C at $t = 1$. Your final answer should be an equation of a line in *slope-intercept form*: $y = mx + b$.
- (b.) What is the equation of the normal line to C at $t = 1$?
- (c.) Sketch the curve, the tangent line, and the normal line. (Please label them).
- (d.) Compute the value of $\frac{d^2y}{dx^2}$ at the point $t = 1$.

- (e.) Based on your answer in part (d.), is the curve C concave up or concave down at $t = 1$? (*Note: You can get the answer by looking at the graph, but the point of this part is that you know how to interpret part (d.)*)

Problem 7 (Differentiating parametric equations). Consider the curve C given by the parametric equations

$$\begin{cases} x = \frac{t}{1+t} \\ y = \sqrt{1+t} \end{cases}$$

for $0 \leq t < \infty$.

- (a.) Compute $\frac{dy}{dx}$.
- (b.) Let $f(t) = \frac{dy}{dx}(t)$. In your words, provide a geometric interpretation or explanation of the function f .
- (c.) Compute the second derivative $\frac{d^2y}{dx^2}$.
- (d.) Based on your answer in part (c.), at the point $t = 1$, is the curve C concave up, concave down, or neither? Justify your answer.